7.2 Answers: column methods and long division

← Unit 7.2 · Practice

The strategy notes describe the thinking, not the layout: follow the written layout he uses at school.

Set 1 · Warm-up: columns with whole numbers

1. 468 (no exchanging needed)
2. 335
3. 722 (7 + 5 = 12, write 2, carry 1; then 8 + 3 + 1 = 12, write 2, carry 1; then 4 + 2 + 1 = 7)
4. 169 (exchange a ten and a hundred; check: 169 + 263 = 432)
5. 4 125
6. 3 766 (check by adding: 3 766 + 1 234 = 5 000)
7. 21 110
8. 25 527 (check: 25 527 + 4 678 = 30 205)
9. 1 000

Set 2 · Building: kommagetallen in columns

1. 3,8
2. 6,85
3. 4,3
4. 16,15 (write 3,7 as 3,70 so the commas line up)
5. 6,55
6. 16,85 (write 25,6 as 25,60)
7. 16,05 (12 is 12,00)
8. a. € 7,44 b. € 2,56 (count up from € 7,44: 6 cents to € 7,50, 50 cents to € 8, € 2 to € 10)
9. 7,35 km
10. 3

Talk about it: 12 is a whole number, so the comma is “invisible” after the 2: 12,00.

Set 3 · Building: long multiplication

1. 63
2. 2 144
3. 2 156 (7 × 300 = 2 100 and 7 × 8 = 56)
4. 180 (10 × 15 = 150, 2 × 15 = 30)
5. 1 058 (3 × 46 = 138, 20 × 46 = 920; estimate 20 × 50 = 1 000)
6. 1 768 (2 × 34 = 68, 50 × 34 = 1 700)
7. 3 000 (4 × 125 = 500, so 24 × 125 = 6 × 500)
8. 1 776 (7 × 48 = 336, 30 × 48 = 1 440; estimate 50 × 40 = 2 000)
9. € 392 (28 × 14)
10. € 345 (15 × 23)
11. 460

Set 4 · Dividing with happy numbers

For each one, check by multiplying back.

1. 21 (80 : 4 = 20, 4 : 4 = 1)
2. 32 (90 : 3 = 30, 6 : 3 = 2)
3. 212
4. 125 (500 : 5 = 100, 125 : 5 = 25)
5. 312
6. 112 (bites: 100 × 6 = 600, 10 × 6 = 60, 2 × 6 = 12)
7. 208 (200 × 7 = 1 400, leaves 56, 8 × 7 = 56)
8. 282 (check: 282 × 9 = 2 538)
9. 64 (60 × 12 = 720, leaves 48, 4 × 12 = 48)
10. 45 (40 × 25 = 1 000, leaves 125, 5 × 25 = 125)
11. 100

Set 5 · Stretch: remainders and kommagetallen

1. 6 rest 1
2. 9 rest 2
3. 91 rest 7 (9 × 91 = 819, 826 − 819 = 7)
4. 25,2 (25 rest 1, and 1 : 5 = 0,2)
5. 12,5 (12 rest 2, and 2 : 4 = 0,5)
6. 125
7. 3 buses (130 : 50 = 2 rest 30; the 30 children need a bus too, so round up)
8. € 7,50 (30 : 4 = 7 rest 2, and € 2 : 4 = € 0,50)
9. 16 full boxes, 4 eggs left over (16 × 6 = 96)
10. 1,25 m (10 : 8 = 1 rest 2; 2 m : 8 = 0,25 m; or 1 000 cm : 8 = 125 cm)
11. 30

Talk about it: buses round up (everyone must go), money keeps dividing into cents, egg boxes count only full ones and say what's left.

Puzzle corner

- **Digit cards:** biggest is 41 × 32 = 1 312 (just ahead of 42 × 31 = 1 302). The big digits go in the tens places; then it's worth trying both ways. Smallest is 13 × 24 = 312 (small digits in the tens places, and again try both ways: 14 × 23 = 322).
- **Remainder detective:** 13, 25, 37 and 49 (and 1, if you allow 1 : 2 = 0 rest 1). They go up in steps of 12, because 12 is the smallest number that 2, 3 and 4 all divide into.